CSE490 (D)
Introduction to
Football Data & Analytics
Lecture-1

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Course Objectives

- Introduce research-oriented football analytics using computing and statistical methods
- Develop analytical, modeling, and visualization skills using real-world data
- Foster independent research, critical thinking, and lifelong learning
- Strengthen ethical reasoning, teamwork, and professional communication

Course Content

Module	Lessons
Data Foundations & Visualization	Introduction to Event Data & KPIs; Visualizing Football; Passing Networks & Team Structure
Statistical Models & Expected Goals	Choosing & Fitting Models; Advanced Models; Building an xG Model
Player & Action Valuation	Scouting and Player Profiles; Valuing Specific Actions
Uncertainty, Prediction & Trends	Randomness in Football; Team-Level KPIs & Trend Analysis
Space Control & Player Movement	Pitch Control Tools; Off-Ball Movement & Positioning
Physical Data & Clustering Techniques	Physical Performance Metrics; Clustering in Football Analytics

Assessment Structure

Assessment	Weight
Attendance Quiz	5% 10%
Midterm	20%
Research Project	25%
Mini Case Study	10%
Final	20%
Research Presentation	10%

Research Project

Nature

- Group (4 students)
- Dataset: StatsBomb / Wyscout / Open football data

Deliverables

- ❖ Research proposal
- ❖ Literature review
- ❖ Methodology & modeling
- ❖ Results & discussion
- ❖ Ethical considerations
- ❖ Final report & presentation

Research Project Structure

Assessment	Weight
Problem Formulation	15%
Literature Review	15%
Methodology	20%
Analysis & Validation	25%
Ethics & Limitation	10%
Communication	15%

Tools

- Python
- Open datasets: StatsBomb / WyScout
- Libraries: pandas, numpy, matplotlib, mplsoccer, statsbombpy

- ***N:B:*** We will mostly use google colab, but it's better if you have conda/ VScode installed

Statsbomb (Data structure)

Follow this notebook- https://colab.research.google.com/drive/11izIsskxXn91-AVLfADYO0GOR1sFivDd?usp=sharing

Data Structure: *Competition → Season → Match → Events*

How to access data:

- ❑ StatsBomb assigns unique IDs at each level
- ❑ Each competition, season, and match has its own ID
- ❑ Example
  - Competition: FIFA World Cup, competition_id = 43
  - Season: 2018, season_id = 106
  - Match: England vs Colombia, match_id = 7580

Statsbomb (Competitions)

Loading competitions as a DataFrame:

- Provides high-level information such as:
  - Competition name / country
  - Season
  - Number of matches
  - Gender
  - Availability of **360 data**

#	Column	Non-Null Count	Dtype
0	competition_id	75 non-null	int64
1	season_id	75 non-null	int64
2	country_name	75 non-null	object
3	competition_name	75 non-null	object
4	competition_gender	75 non-null	object
5	competition_youth	75 non-null	bool
6	competition_international	75 non-null	bool
7	season_name	75 non-null	object
8	match_updated	75 non-null	object
9	match_updated_360	57 non-null	object
10	match_available_360	11 non-null	object
11	match_available	75 non-null	object

Statsbomb (Matches)

Loading matches for a competition as a DataFrame:

- Provides match-level information such as:
  - Home team & away team
  - Match score
  - Stadium
  - Match stage
  - Referee
  - Team managers
  - Match date

#	Column	Non-Null Count	Dtype
0	match_id	64 non-null	int64
1	match_date	64 non-null	object
2	kick_off	64 non-null	object
3	competition	64 non-null	object
4	season	64 non-null	object
5	home_team	64 non-null	object
6	away_team	64 non-null	object
7	home_score	64 non-null	int64
8	away_score	64 non-null	int64
9	match_status	64 non-null	object
10	match_status_360	64 non-null	object
11	last_updated	64 non-null	object
12	last_updated_360	64 non-null	object
13	match_week	64 non-null	int64
14	competition_stage	64 non-null	object
15	stadium	64 non-null	object
16	referee	64 non-null	object
17	home_managers	64 non-null	object
18	away_managers	64 non-null	object
19	data_version	64 non-null	object
20	shot_fidelity_version	0 non-null	object
21	xy_fidelity_version	0 non-null	object

Statsbomb (Events)

Loading events for a match as a DataFrame:

- Each row represents **one on-ball action**
- Examples of events:
  - Pass
  - Shot
  - Foul
  - Duel
  - Clearance

Information stored in event data includes:

- Event type
- Time (minute, second)
- Event location on the pitch
- Player and team involved
- Event-specific details (e.g., pass type, shot outcome)

Statsbomb (Events)

Important observation:

- Most columns contain **NaN values**
- This is expected because:
  - Each event type only uses a **subset of columns**
  - Columns not relevant to an event remain empty

```
Data columns (total 83 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   50_50                         4 non-null      object
 1   bad_behaviour_card            3 non-null      object
 2   ball_receipt_outcome          160 non-null    object
 3   ball_recovery_recovery_failure 13 non-null    object
 4   block_deflection              2 non-null      object
 5   block_offensive               1 non-null      object
 6   carry_end_location            851 non-null    object
 7   clearance_aerial_won          4 non-null      object
 8   counterpress                  108 non-null    object
 9   dribble_nutmeg                2 non-null      object
 10  dribble_outcome               29 non-null     object
 11  duel_outcome                  27 non-null     object
 12  duel_type                     62 non-null     object
 13  duration                      2411 non-null   float64
 14  foul_committed_advantage      2 non-null      object
 15  foul_committed_card           5 non-null      object
 16  foul_committed_penalty        1 non-null      object
```

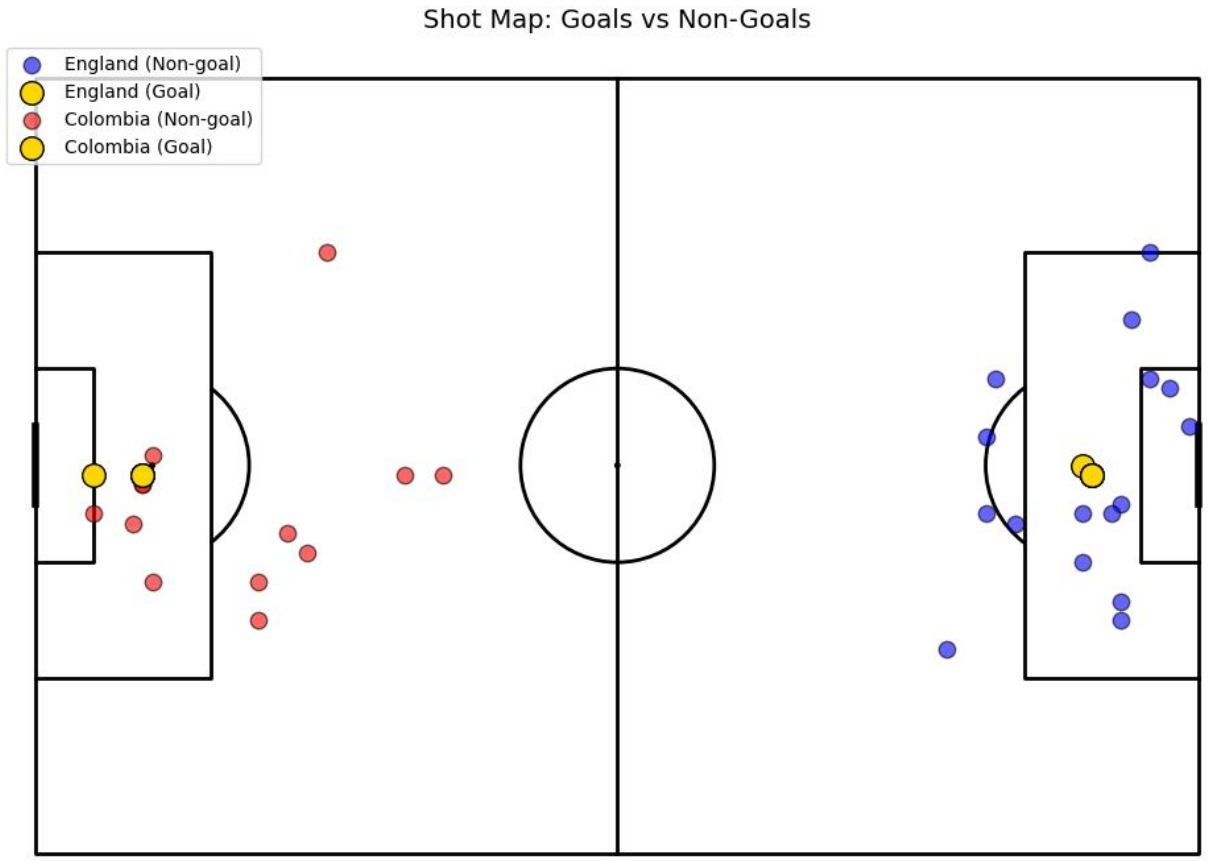
Statsbomb (Events= Shots)

We can extract very specific info (Non-NaN) of a “shot” event

Event Type	Player	Team	Minute	Location (x, y)	Body Part	Shot Type	Outcome	xG
Shot	Ashley Young	England	5	(115.0, 18.0)	Right Foot	Free Kick	Saved	0.0098
Shot	Raheem Sterling	England	7	(112.0, 54.0)	Right Foot	Open Play	Blocked	0.0382
Shot	Raheem Sterling	England	12	(98.0, 37.0)	Right Foot	Open Play	Blocked	0.0451
Shot	Harry Kane	England	15	(119.0, 36.0)	Head	Open Play	Off Target	0.6251
Shot	Juan Cuadrado	Colombia	21	(97.0, 56.0)	Right Foot	Open Play	Blocked	0.0218

Statsbomb (Visualizing shots)

We can draw standard size football pitch and visualize events using **mplsoccer**'s **Pitch**



Statsbomb (360 Data)

Statsbomb offers 360 data which track not only location of an event but also players' location.

#	Column	Non-Null Count	Dtype
0	id	41562 non-null	object
1	visible_area	41562 non-null	object
2	match_id	41562 non-null	int64
3	teammate	41562 non-null	bool
4	actor	41562 non-null	bool
5	keeper	41562 non-null	bool
6	location	41562 non-null	object

360 frame info

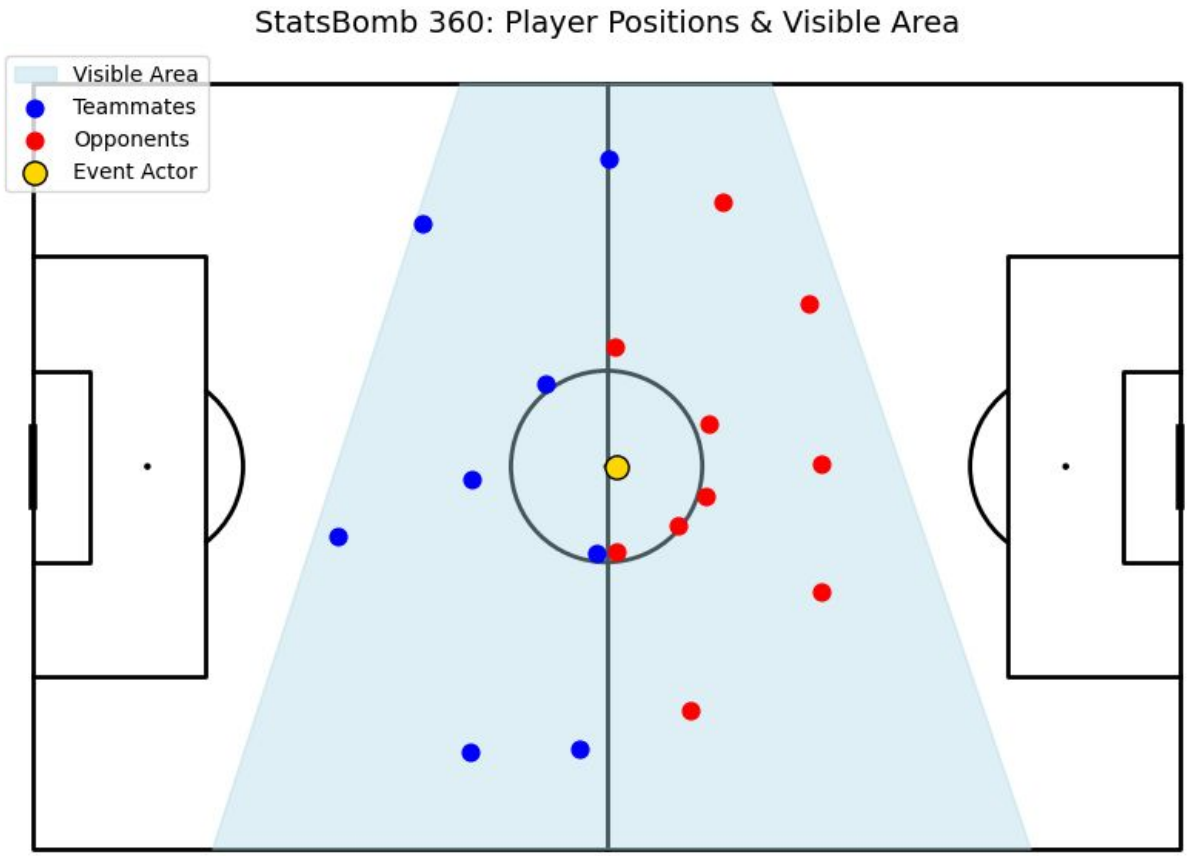
Player locations at the moment of the event:

	x	y	teammate	actor	keeper
0	31.885449	47.289130	True	False	False
1	40.762566	14.657225	True	False	False
2	45.765232	69.890950	True	False	False
3	45.923999	41.324219	True	False	False
4	53.649592	31.368630	True	False	False
5	57.221077	69.484720	True	False	False
6	58.963536	49.056717	True	False	False
7	60.159949	7.849366	True	False	False
8	60.952056	27.553933	False	False	False

Player locations at the moment of a particular event

Statsbomb (360 Data)

What can you observe from this pitch image?



Serbia vs Switzerland
Match date: 2022-12-02
Serbia starting formation: 3412
Switzerland starting formation: 4231
These are also extracted by using match_id and keys like home_team, away_team, Match_date, type, teactics

Event Data vs Tracking Data

Event Data

- Records **on-ball actions** (passes, shots, tackles, fouls)
- Logged **only when an event occurs**
- Sparse data, many **NaN columns** (not all fields apply to all events)
- Used for **xG, shot maps, pass maps, KPIs**

Tracking Data

- Records **continuous positions** of all players & the ball
- Collected **every frame** (e.g, 10 times/sec)
- Dense, large datasets
- Used for **pressing, team shape, space control, movement analysis**

References

- **Soccermatics Blog:** Welcome to Soccermatics — Soccermatics documentation
- Course: mathematical modelling of football - YouTube
- **Book: Soccermatics by David Sumpter**